

Chaithanya Naik

Member of Technical Staff
R&D - Avi Networks
VMware Inc.

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in chaithanya-naik

Research Interests

Quantum Computing and Information, Quantum Control, Quantum algorithms, Machine learning and AI

Education

Indian Institute of technology Bombay 2020-2021

M.Tech in Computer Science and Engineering GPA (thesis) : 10.0/10.0

Advisor: Prof. Sai Vinjanampathy

Indian Institute of technology Bombay 2016-2020

B.Tech in Computer Science and Engineering GPA: 7.25/10.0

Minor in Physics

Minor in AI and Data Science

Advisor: Prof. Amit Sethi

Research Experience

Robust Quantum Optimal Control Spring 2021

Guide: Prof. Sai Vinjanampathy

Master's Thesis Project, IIT Bombay

Submitted abstract to APS March meeting 2022

- Designed sequence-to-sequence learning model using deep networks based on encoder-decoder architecture consisting of CRNNs to generate the optimized control sequences for transmon and cross resonance systems despite noise
- Enhanced the performance of quantum gate compilations by 100x compared to current state of the art solutions
- Analyzed literature on Geodesic analysis, Deep Sequential models, and Optimal Quantum Control methods

Controller Synthesis Spring 2021

Guide: Prof. Ashutosh Gupta

R&D Project, IIT Bombay

- Synthesised a controller for real-time system, based on data-driven RL and algorithmic SMT/SAT approaches to control a railway network modelled as a network of timed automata constrained over a set of specifications
- Utilised tools like UPPAAL and DCValid to design and model networks of timed automata and verify solutions
- Studied about determinization and minimization of timed automata for a specification given in duration calculus

Modular Quantum Computer Design Autumn 2020

Guide: Prof. Sai Vinjanampathy

Master's Thesis Project, IIT Bombay

- Designed robust control protocols and generalized for n-Qubit Gates to enable fault-tolerant quantum computation
- Implemented a modular RL framework for determining the optimal pulse sequence for any general quantum gate preparation using Deep RL through DQN, DDPG, Trust Region Policy Optimization, and Actor-Critic methods

Interactive Image Segmentation Autumn 2020

Guide: Prof. Amit Sethi

Bachelor's Thesis Project, IIT Bombay

- Automated the generation of markers required to segment desired object in image with minimal user intervention
- Designed a deep network using prior based feature enrichment network (PFE-net) and feature back-propagating refinement scheme (f-BRS) on PASCAL-5i dataset, to decrease the number of user clicks required
- Experimented with SeedNet, a DQN approach over MSRA10k dataset to achieve IOU of around 60.5

Quantum Gate Optimization Autumn 2019

Guide: Prof. Sai Vinjanampathy

R&D Project, IIT Bombay

- Implemented Subspace Selective Self-Adaptive Differential Evolution (SUSSADE) technique to generate the instructions for effectively determining the control parameters of the quantum control scheme
- Critically analyzed advantages of GrAPE with Push-Pull optimization and RL approaches like Q-learning, Deep-Deterministic Policy gradient methods and evolutionary RL for improving existing bounds for gate fidelity

Professional Experience

VMware Inc.

July 2021 - present

Avi NSX-T load-balancer | Mentor: Anurag Palsule

Bangalore, India

- Working on enhancing the performance of load balancer to manage heavy workloads through kubernetes operator
- Part of the team maintaining an open source project, catering to needs for NSX-T and openshift platforms
- Worked on functional test-suites to ensure better performance for individual components

Amazon Development Centre India Pvt. Ltd

May - July 2019

Auto Trouble Ticket Manager | Mentor: Archit Agrawal

Hyderabad, India

- Automated process of resolving trouble tickets raised on amazon services by developing java package
- Designed runtime compilation module using Java Compiler API to implement dynamic code execution by converting java snippet code from different source packages into a compiled java object
- Developed a generic framework to plug-in their models for information retrieval and auto-suggesting resolution

Awards & Achievements

- Secured 1st position in ISRO's web visualization tool for Astrosat data at 9th Inter IIT Tech Meet 2021
- Bagged Gold Medal in Bosch's Route Optimization challenge at the 8th Inter IIT Tech Meet 2019
- Secured 2nd position in Capture The Flag cybersecurity challenge at the 8th Inter IIT Tech Meet 2019
- Bagged Gold Medal at the 7th Inter IIT Tech Meet in Star Cluster Identifier competition 2018
- Secured AIR 340 in IIT-JEE Advanced | AIR 176 in JEE Main Paper-I | AIR 90 in JEE Main Paper-II 2016
- Amongst Top 300 students in the country qualified for INPhO (for two straight years) and INMO 2015,14
- Achieved KVPY Fellowship with All India Rank 173 organized under DST, Government of India 2015
- Recipient of NTSE Scholarship awarded by NCERT under Ministry of HRD, Government of India 2012

Teaching & Mentoring Experience

Teaching Assistant

July 2020 - June 2021

- Responsible for clarifying queries of students, helping with course material and grading assignments
- Courses Taught - Quantum Computing and Information (Spring 2021) and python programming lab (Autumn 2020)

Mentor

Quantum Computing, Computer Vision: Maths and Physics Club

April - June 2020

- Spoof-Resistant Facial Recognition using Deep Learning - guided a group of 4 students
- Designing Rubik's Cube Solver using Reinforcement Learning using DeepCubeA - guided a group of 6 students
- Project on creating star hopping guide & creating astronomy website using Django - guided a group of 6 students

Positions of Responsibility

Department General Secretary | CSE Department, IIT Bombay

April 2019 - July 2020

- Elected student representative in department policy formulation committees to ensure student involvement
- Re-instantiated and managed Cyber-Security Club, organized CTF events to increase awareness about cybersecurity
- Spearheaded a 3-tier team of 12 members in designing and development of the content for department newsletter

Department Academic Mentor | CSE Department, IIT Bombay

June 2020 - July 2021

- Mentoring 4 sophomores for their academic and general concerns, and helping them cope with the curriculum

Technical Skills

Programming

C/C++, Python, Java, R, Racket(Scheme), SWI-Prolog, Bash, VHDL

Deep Learning

PyTorch, TensorFlow, Keras, TensorFlow Quantum

Extracurriculars

- Part of the Inter IIT contingent securing Runner's Up Position at Inter IIT Tech Meet held at IIT Bombay 2018
- Awarded Hostel Player of GC (Carrom) for best performance in Inter Hostel Sports, IIT Bombay 2016
- Completed a year-long course in Athletics offered by National Sports Organization (NSO), IIT Bombay 2016
- Attended Vijyoshi Science Camp organized by Indian Institute of Science (IISc), Bengaluru, India 2015